

CPU Scheduling Simulator — C5

Round Robin vs Shortest Job First (SJF)

Project Description :

This document presents the implementation and comparison of three CPU scheduling algorithms: Round Robin (RR), Shortest Job First (SJF), and Shortest Remaining Time First (SRTF). The project compares all three algorithms using the same workloads and evaluates their performance using scheduling metrics such as Waiting Time (WT), Turnaround Time (TAT), and Response Time (RT).

Team Information :

Name	Id
زياد محمد محمد الراوي سعيد	20240389
خالد احمد حمدي	20240321
زياد عمرو جمال	20240382
شادي عبده محمد عبده	20240471
علي محمد عبدالسلام عبدالصمد	20240603
عمر محمد اسامه عبيد	20240638

Project :

Round Robin (RR):

Round Robin distributes CPU time fairly between processes using a fixed time quantum. Each process receives a small portion of CPU time before the scheduler switches to the next process.

Shortest Job First (SJF):

SJF selects the process with the shortest burst time first. It improves efficiency and usually reduces average waiting time

Shortest Remaining Time First (SRTF):

SRTF is the preemptive version of SJF. It always selects the process with the shortest remaining burst time and can interrupt a running process if a shorter job arrives. This improves response time and reduces waiting time for short processes.

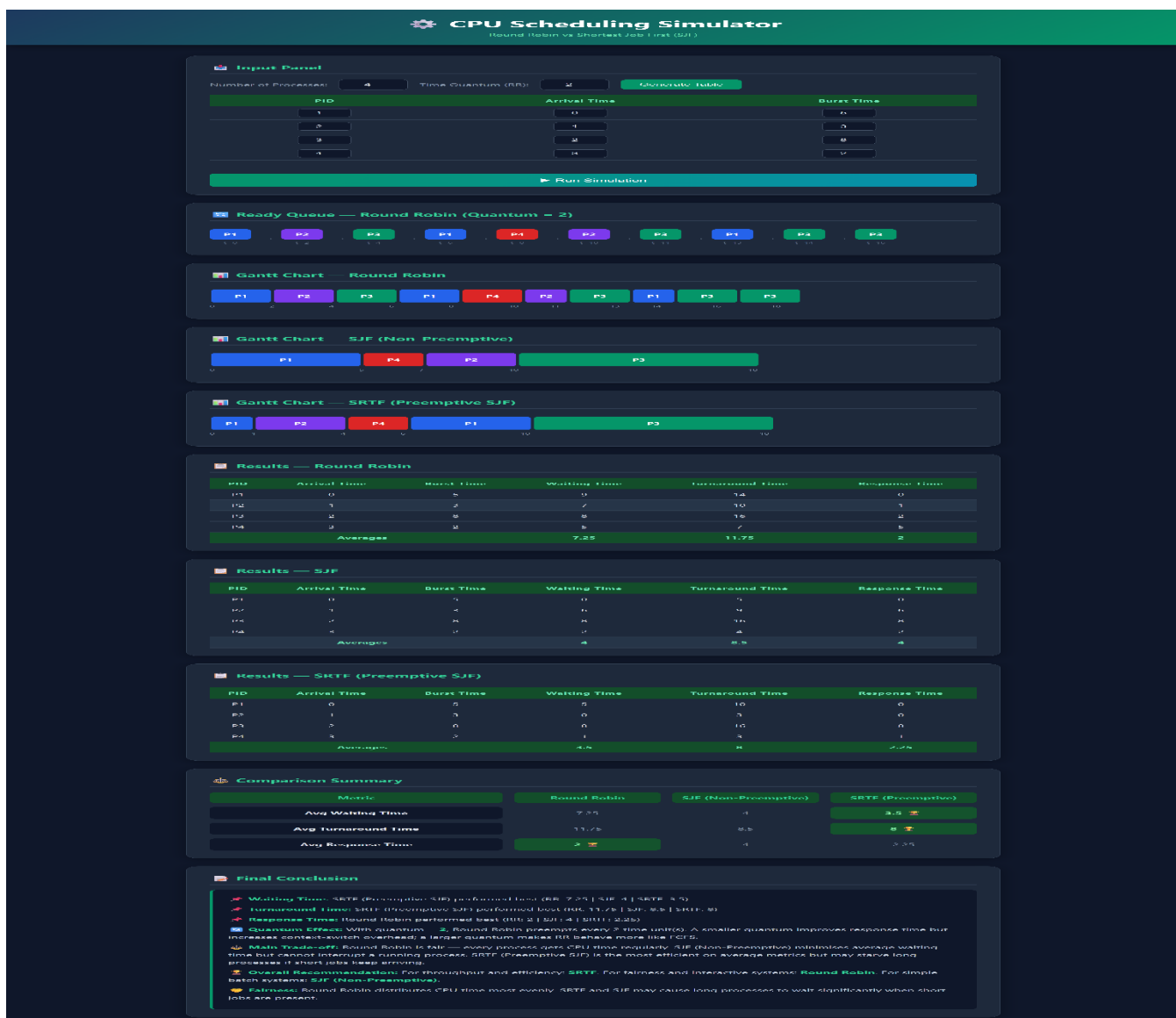
Metrics

- Waiting Time (WT) = Turnaround Time - Burst Time
- Turnaround Time (TAT) = Completion Time - Arrival Time
- Response Time (RT) = First Start Time - Arrival Time

Scenarios

Scenario A — Basic Mixed Workload :

Goal: A normal workload with different arrival times and burst times to establish baseline comparison between RR and SJF.



Scenario B — Short-Job-Heavy

Goal: Many short jobs so SJF behavior becomes very visible and clearly favors short burst times.

CPU Scheduling Simulator

Round Robin vs Shortest Job First (SJF)

Input Panel

Number of Processes: 5

Time Quantum (RR): 2

Generate Table

PID	Arrival Time	Burst Time
1	0	1
2	0	2
3	0	1
4	0	3
5	0	2

Run Simulation

Ready Queue — Round Robin (Quantum = 2)

P1

P2

P3

P4

P5

P4

Gantt Chart — Round Robin

P1

P2

P3

P4

P5

P4

Gantt Chart — SJF (Non-Preemptive)

P1

P3

P2

P5

P4

Gantt Chart — SRTF (Preemptive SJF)

P1

P3

P2

P5

P4

Results — Round Robin

PID	Arrival Time	Burst Time	Waiting Time	Turnaround Time	Response Time
P1	0	1	0	1	0
P2	0	2	1	3	1
P3	0	1	1	2	0
P4	0	3	5	8	4
P5	0	2	6	8	6
Averages			3.2	5	2.8

Results — SJF

PID	Arrival Time	Burst Time	Waiting Time	Turnaround Time	Response Time
P1	0	1	0	1	0
P2	0	2	2	4	2
P3	0	1	1	2	0
P4	0	3	6	9	6
P5	0	2	4	6	4
Averages			2.6	4.4	2.6

Results — SRTF (Preemptive SJF)

PID	Arrival Time	Burst Time	Waiting Time	Turnaround Time	Response Time
P1	0	1	0	1	0
P2	0	2	2	4	2
P3	0	1	1	2	0
P4	0	3	6	9	6
P5	0	2	4	6	4
Averages			2.6	4.4	2.6

Comparison Summary

Metric	Round Robin	SJF (Non-Preemptive)	SRTF (Preemptive)
Avg Waiting Time	3.2	2.6	2.6
Avg Turnaround Time	5	4.4	4.4
Avg Response Time	2.8	2.6	2.6

Final Conclusion

Waiting Time: SRTF (Preemptive SJF) performed best (RR: 3.2 | SJF: 2.6 | SRTF: 2.6)

Turnaround Time: SRTF (Preemptive SJF) performed best (RR: 5 | SJF: 4.4 | SRTF: 4.4)

Response Time: SRTF (Preemptive SJF) performed best (RR: 2.8 | SJF: 2.6 | SRTF: 2.6)

Quantum Effect: With quantum = 2, Round Robin preempts every 2 time units. A smaller quantum improves response time but increases context switch overhead; a larger quantum makes RR behave more like FCFS.

Main Trade-off: Round Robin is fair — every process gets CPU time regularly. SJF (Non-Preemptive) minimises average waiting time but cannot interrupt a running process. SRTF (Preemptive SJF) is the most efficient on average metrics but may starve long processes if short jobs keep arriving.

Overall Recommendation: For throughput and efficiency: SRTF. For fairness and interactive systems: Round Robin. For simple batch systems: SJF (Non-Preemptive).

Fairness: Round Robin distributes CPU time most evenly. SRTF and SJF may cause long processes to wait significantly when short jobs are present.

Goal: Show how Round Robin distributes CPU time more fairly across all processes compared to SJF

CPU Scheduling Simulator

Round Robin vs Shortest Job First (SJF)

Input Panel

Number of Processes: 3Time Quantum (RR): 3Generate Table

PID	Arrival Time	Burst Time
1	0	10
2	0	10
3	0	10

Run Simulation

Ready Queue — Round Robin (Quantum = 3)

P1P2P3P1P2P3P1P2P3P1P2

t=0t=1t=6t=9t=12t=15t=18t=21t=24t=27t=29t=30

Gantt Chart — Round Robin

P1P2P3P1P2P3P1P2P3P1P2P3

03691215182124272930

Gantt Chart — SJF (Non-Preemptive)

P1P2P3

0102030

Gantt Chart — SRTF (Preemptive SJF)

P1P2P3

0102030

Results — Round Robin

PID	Arrival Time	Burst Time	Waiting Time	Turnaround Time	Response Time
P1	0	10	19	29	0
P2	0	10	19	29	3
P3	0	10	20	30	6
Averages			19	29	3

Results — SJF

PID	Arrival Time	Burst Time	Waiting Time	Turnaround Time	Response Time
P1	0	10	0	10	0
P2	0	10	10	20	10
P3	0	10	20	30	20
Averages			10	20	10

Results — SRTF (Preemptive SJF)

PID	Arrival Time	Burst Time	Waiting Time	Turnaround Time	Response Time
P1	0	10	0	10	0
P2	0	10	10	20	10
P3	0	10	20	30	20
Averages			10	20	10

Comparison Summary

Metric	Round Robin	SJF (Non-Preemptive)	SRTF (Preemptive)
Avg Waiting Time	19	10 🏆	10 🏆
Avg Turnaround Time	29	20 🏆	20 🏆
Avg Response Time	3 🏆	10	10

Final Conclusion

🚀 **Waiting Time:** SRTF (Preemptive SJF) performed best (RR: 19 | SJF: 10 | SRTF: 10).

🚀 **Turnaround Time:** SRTF (Preemptive SJF) performed best (RR: 29 | SJF: 20 | SRTF: 20).

🚀 **Response Time:** Round Robin performed best (RR: 3 | SJF: 10 | SRTF: 10).

💡 **Quantum Effect:** With quantum = 3, Round Robin preempts every 3 time units(s). A smaller quantum improves response time but increases context switch overhead; a larger quantum makes RR behave more like FCFS.

⚖️ **Main Trade-off:** Round Robin is fair — every process gets CPU time regularly. SJF (Non-Preemptive) minimises average waiting time but cannot interrupt a running process. SRTF (Preemptive SJF) is the most efficient on average metrics but may starve long processes if short jobs keep arriving.

📊 **Overall Recommendation:** For throughput and efficiency: **SRTF**. For fairness and interactive systems: **Round Robin**. For simple batch systems: **SJF (Non-Preemptive)**.

❤️ **Fairness:** Round Robin distributes CPU time most evenly. SRTF and SJF may cause long processes to wait significantly when short jobs are present.

Scenario D — Long-Job Sensitivity

Goal: One long process competes with shorter ones. SJF (non-preemptive) cannot interrupt it, causing unfair delay for short jobs that arrive later.

CPU Scheduling Simulator

Round Robin vs Shortest Job First (SJF)

Input Panel

Number of Processes:3Time Quantum (RR):4Generate Table

PID	Arrival Time	Burst Time
1	0	15
2	2	2
3	4	3

Run Simulation

Ready Queue — Round Robin (Quantum = 4)

P1P2P3P1P1P1

0.00.40.81.21.61.7

Gantt Chart — Round Robin

P1P2P3P1P1P1

0468121620

Gantt Chart — SJF (Non-Preemptive)

P1P2P3

0151720

Gantt Chart — SRTF (Preemptive SJF)

P1P2P3P1

024720

Results — Round Robin

PID	Arrival Time	Burst Time	Waiting Time	Turnaround Time	Response Time
P1	0	15	5	20	0
P2	2	2	2	4	2
P3	4	3	2	5	2
Averages			3	9.67	1.33

Results — SJF

PID	Arrival Time	Burst Time	Waiting Time	Turnaround Time	Response Time
P1	0	15	0	15	0
P2	2	2	13	15	13
P3	4	3	13	16	13
Averages			8.67	15.33	8.67

Results — SRTF (Preemptive SJF)

PID	Arrival Time	Burst Time	Waiting Time	Turnaround Time	Response Time
P1	0	15	5	20	0
P2	2	2	0	2	0
P3	4	3	0	3	0
Averages			1.67	8.33	0

Comparison Summary

Metric	Round Robin	SJF (Non-Preemptive)	SRTF (Preemptive)
Avg Waiting Time	3	8.67	1.67
Avg Turnaround Time	9.67	15.33	8.33
Avg Response Time	1.33	8.67	0

Final Conclusion

⚡ Waiting Time: SRTF (Preemptive SJF) performed best (RR: 3 | SJF: 8.67 | SRTF: 1.67)

🔥 Turnaround Time: SRTF (Preemptive SJF) performed best (RR: 9.67 | SJF: 15.33 | SRTF: 8.33)

📊 Response Time: SRTF (Preemptive SJF) performed best (RR: 1.33 | SJF: 8.67 | SRTF: 0)

🔄 Quantum Effect: With quantum = 4, Round Robin preempts every 4 time unit(s). A smaller quantum improves response time but increases context switch overhead; a larger quantum makes RR behave more like FCFS.

⚖️ Main Trade-off: Round Robin is fair — every process gets CPU time regularly. SJF (Non-Preemptive) minimises average waiting time but cannot interrupt a running process. SRTF (Preemptive SJF) is the most efficient on average metrics but may starve long processes if short jobs keep arriving.

🏆 Overall Recommendation: For throughput and efficiency: SRTF. For fairness and interactive systems: Round Robin. For simple batch systems: SJF (Non-Preemptive).

👉 Fairness: Round Robin distributes CPU time most evenly. SRTF and SJF may cause long processes to wait significantly when short jobs are present.

Scenario E — Validation Case

Input Panel

Number of Processes:

3

Time Quantum (RR):

2

Generate Table

PID	Arrival Time	Burst Time
1	-1	15
2	-1	2
3	-1	3

✗ Process 1: Arrival Time must be ≥ 0 .

▶ Run Simulation

Input Panel

Number of Processes:

3

Time Quantum (RR):

1

Generate Table

PID	Arrival Time	Burst Time
1	0	require
2	0	require
3	0	require

✗ Process 1: Burst Time must be > 0 .

▶ Run Simulation

Input Panel

Number of Processes:

3

Time Quantum (RR):

2

Generate Table

PID	Arrival Time	Burst Time
1	0	0
2	0	0
3	0	0

✗ Process 1: Burst Time must be > 0.

▶ Run Simulation

Input Panel

Number of Processes:

3

Time Quantum (RR):

2

Generate Table

PID	Arrival Time	Burst Time
1	0	15
1	0	2
3	0	3

✗ Duplicate PID found: 1

▶ Run Simulation

Input Panel

Number of Processes:

3

Time Quantum (RR):

0

Generate Table

PID	Arrival Time	Burst Time
1	0	15
2	0	2
3	0	3

✖ Time Quantum must be a positive integer.

▶ Run Simulation